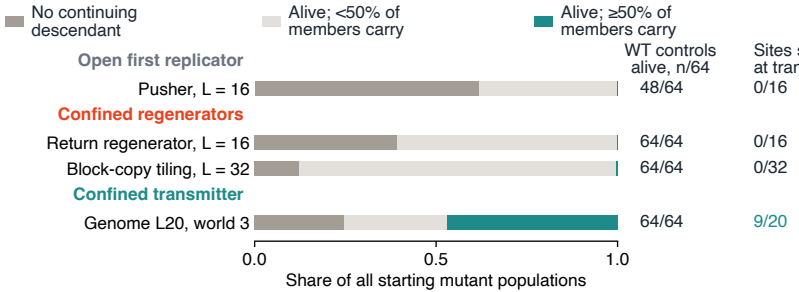
a **Carriage can fail despite survival (generation 8)**

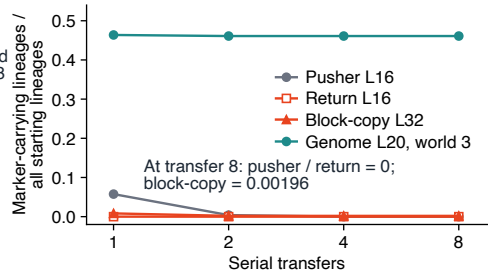


L = tape length in bytes. Alive: ≥75% parent-byte match; carrier: retains mutant 5-byte window

Site (a,c): mutant carrier fraction – WT carrier fraction ≥0.5 for ≥half identifiable byte values

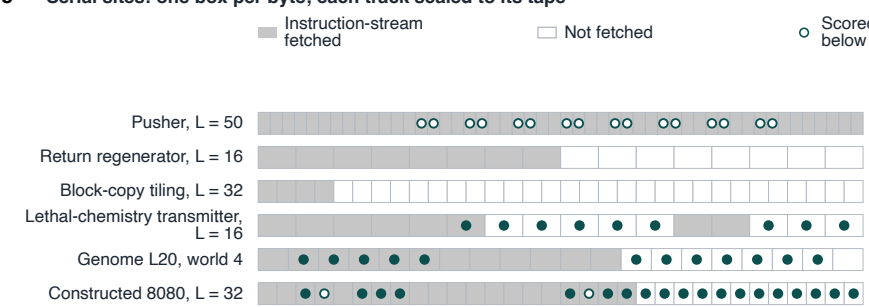
Branching mutant starts: L16 2,048; L32 4,096; L20 2,560; 64 WT (unmutated) controls/design

b **Serial retention (historical)**



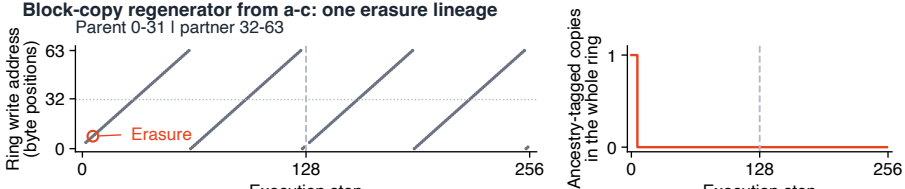
Serial: one member; carrier retains mutant 5-byte window
255 values/site × 16 starts/value × L
256 unmutated controls

c **Serial sites: one box per byte; each track scaled to its tape**



	Fetched bytes	Scored sites at transfer 1	Scored sites at transfer 8	Of these: not fetched
Pusher, L = 50	50/50	16	0	0
Return regenerator, L = 16	8/16	0	0	0
Block-copy tiling, L = 32	4/32	0	0	0
Lethal-chemistry transmitter, L = 16	8/16	9	9	8
Genome L20, world 4	12/20	12	12	7
Constructed 8080, L = 32	20/32	21	19	12

d **Block-copy regenerator from a-c: one erasure lineage**



Ancestry label: byte 8 (0xA7)

Erasure at execution step 6

Read address 4; ancestry: byte 0
Writes 0x04 over byte 8 (0xA7)

Labelled byte copies: 1 to 0

128: offspring to parent; CPU reset